

2023

Bachelor in Business Administration

Course Title: **Database Management System**

Code No: **N/A** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 1 = 10]

1. Define Data Model.
2. Write SQL statement to delete student table.
3. Differentiate between simple composite attribute.
4. List the types of mapping constraints in ER Model.
5. Write a syntax for check constraint SQL.
6. Define Transaction.
7. What is NoSQL?
8. Define big data.
9. State 1NF.
10. Why level of abstraction is maintained in database?

Section B

Short Answer Questions:(Attempt any FIVE Questions)[5 × 3 = 15]

11. Describe two phase locking mechanism in brief.
12. What are the types of database user? Explain each of them in brief.
13. Explain briefly the importance and methods of Database recovery.
14. Explain the importance of Creating view with syntax.
15. What is Database security? How Database is secured? Explain in brief.
16. What is importance of normalization? Explain with example.

Section C

Long Answer Questions:(Attempt any THREE Questions)[3 × 5 = 15]

17. Identify and explain different types of database languages with example.
18. Explain the desirable properties of Transactions.
19. Explain any five aggregate function used in SQL with syntax.
20. When generalization and specialization is used in ER model? Support your answer with example.

Section D

Comprehensive Answer / Case / Situation Analysis Questions:[2 × 10 = 20]

21. Employee works on the project under one or more project manager for the company. Client pays for the project that is being developed by the company and client provide the feedback for the project. Draw ER diagram for the above scenario.
22. Consider following relational database: Account(ac_no, type, balance, branch) Customer(cid, name, address, phone_no, dob) Depositer(ac_no, cid, date) Write SQL statement for the following: i) Insert a tuple {1001, checking, 30000} in Account relation. ii) Delete those accounts which haven't owns any customer. iii) Change the phone_no of Ram to 11111111. iv) Find name of customer who owns an account with balance greater than 2000. v) Display those records of customers whose name starts with "R" and end with "A".